

DONALD MCKENDRICK

Product & Technology Leader • Interactive Learning • AI in Education • Game Design

Walnut Creek, CA • ddmckendrick@gmail.com • linkedin.com/in/donaldmckendrick • github.com/ddmck

SUMMARY

Technology executive and product leader with 10+ years building interactive learning experiences at scale. Unique combination of shipped ed-tech products (1.5M+ users), hands-on AI/ML expertise (NLP, deep learning, GenAI), game development experience, and a scientific foundation in computational simulation. Track record of translating ambitious educational visions into production-quality interactive products with small, cross-functional teams. Passionate about the intersection of game design, learning science, and emerging technology.

EXPERIENCE

Closegap

Nov 2024 – Present

Chief Technology Officer

San Francisco Bay Area (Remote)

Previously Director of Technology (May 2023 – Nov 2024)

- Lead product strategy and engineering for free mental health and SEL tools used in K–12 schools nationwide. Run development in six-week Shape Up cycles with dedicated prototyping weeks and cooldown periods.
- Built a GenAI-powered crisis detection tool that replaced a keyword system with context-aware AI, personally writing comprehensive test samples and remaining on PagerDuty for every alert—responsible AI in a high-stakes K–12 context.
- Collaborating with Northeastern University’s IASL lab on an early-stage research study examining emotional granularity in students, grounding product decisions in rigorous research.
- Replaced voice actors with AI-generated dynamic responses that mirror students’ feelings back in age-appropriate tones and multiple languages. Hands-on player-coach: set architecture, write production code, review all output.

Amid Labs

Mar 2024 – Jun 2024

Amid Fellow – 24Δ

San Francisco Bay Area

- Selected for competitive fellowship focused on emerging technology and innovation.

Chan Zuckerberg Initiative

Mar 2020 – Aug 2022

Senior Software Engineer

Redwood City, CA

- Worked on curriculum and differentiation teams of CZI’s project-based learning platform; led the 2.0 launch of Announcements—the primary landing page for millions of users, critical during COVID.
- Introduced Puppeteer-based feature testing infrastructure and onboarded the engineering team. Full-stack: Ruby on Rails (Sorbet), React.

Derivative Daydream Machine

Aug 2022 – Nov 2024

Founder & Game Developer

San Francisco Bay Area

- Building a turn-based RPG inspired by classic JRPGs, with educational mechanics woven into narrative. Placed 4th in a VR Game Jam with an accessible party game featuring progressive difficulty and spatial memory mechanics.
- Exploring games addressing mental health themes in connection with Closegap’s SEL work. Technologies: Unity, 3D simulation, VR.

One Medical

Mar 2019 – Mar 2020

Software Engineer

San Francisco Bay Area

- Data Interoperability team: applied ML to improve admin workflows in a HIPAA-compliant healthcare environment, gaining experience with rigorous data privacy and security practices.

Quill.org

Jan 2016 – Feb 2019

Cofounder & Technical Director

New York City

- Co-founded and scaled an interactive literacy platform from 10K to 1.5M+ users (60K DAU), shipping three core tools: Connect (sentence combining), Lessons (prototype in 1 week, shipped in 8), and Diagnostic (22-question adaptive assessment with differentiated recommendations).
- Introduced AI/NLP in 2016 for real-time feedback on open-ended writing—pioneering an approach now widespread in ed-tech. Built deep learning sentence fragment detector, leveraged Paul Allen Institute NLP tools and spaCy, and developed a reverse sentence completion model for automated grading.
- Partnered with Bruce Saddler’s lab on efficacy study (TOWL pre/post). Curriculum grounded in Writing Revolution/Writing Next. Open source codebase with community contributions.
- Raised \$3M+ (2018–2019) through technical grant writing—concept briefs, learning rationale, evidence-of-impact narratives. Built Twilio SMS prototype for live funder demos.
- Scrum Master: reviewed and deployed all code. Built Jenkins CI/CD pipeline. Stack: Rails, Express, Flask, React, React Native, Angular, PostgreSQL, Firebase, TensorFlow, GraphQL, AWS.

EARLIER CAREER

Search The Sales / Fetch My Fashion / HowAbout

Nov 2014 – Aug 2015

CTO

Edinburgh, UK / NYC

- Led engineering for iOS and web apps. Data enrichment of 1M+ products. Stack: Node, Angular, Rails, Swift.

Float

Aug 2013 – Oct 2014

Developer

Edinburgh, UK

- Built payment systems (Stripe), marketing site, and designed interactive onboarding tutorial for new users.

EDUCATION & RESEARCH

University of Amsterdam

Sep 2011 – May 2013

Assistant Researcher (PhD track – Computational Chemistry)

Amsterdam, Netherlands

- Researched gas absorption in metal-organic frameworks using Monte Carlo and molecular dynamics simulation—designing rigorous computational simulations of complex real-world systems.
- Left PhD to pursue a career in education technology.

Heriot-Watt University

2006 – 2011

Master of Chemistry (MChem)

Edinburgh, UK

- Master’s thesis: quantum simulation of catalytic synthesis of isoquinoline.
- School Officer (2008–2011): student-staff liaison; improved student feedback systems.

SKILLS & TECHNOLOGIES

Product & Leadership: Product strategy & roadmapping, cross-functional team leadership, technical grant writing, stakeholder communication, budget management, Shape Up/Scrum/Agile

AI & Machine Learning: NLP (spaCy, Paul Allen Institute tools), deep learning, GenAI, TensorFlow, scikit-learn, responsible AI in K–12, AI-enabled development (Claude Code)

Game & Interactive: Unity, VR development, 3D simulation, ECS architecture, game mechanics design, music composition, Figma, playtesting, iterative prototyping

Engineering: Ruby on Rails, React, Node/Express, Python/Flask, Angular, Swift, GraphQL, PostgreSQL, Firebase, AWS, GCP, Heroku, Docker, Jenkins, Git

Research & Domain: Research partnerships, efficacy studies, computational simulation (Monte Carlo, MD, QM), K–12 education, PBL, mastery-based learning, SEL

RECOGNITION

The World’s Most Innovative Companies in Education 2018 — Quill.org (Fast Company)

Product Forge 2014 — Winning Team (Education Technology)

Languages: English (native) • German